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**9702/04**

May/June 2007

**1 hour 45 minutes**

No Additional Materials are required.

## READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| 1                  |  |
| 2                  |  |
| 3                  |  |
| 4                  |  |
| 5                  |  |
| 6                  |  |
| 7                  |  |
| 8                  |  |
| 9                  |  |
| 10                 |  |
| 11                 |  |
| Total              |  |

This document consists of **23** printed pages and **1** blank page.

**Data**

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

**Formulae**

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

## Section A

Answer **all** the questions in the spaces provided.

- 1 (a) Explain what is meant by a *gravitational field*.

.....  
 ..... [1]

- (b) A spherical planet has mass  $M$  and radius  $R$ . The planet may be considered to have all its mass concentrated at its centre.  
 A rocket is launched from the surface of the planet such that the rocket moves radially away from the planet. The rocket engines are stopped when the rocket is at a height  $R$  above the surface of the planet, as shown in Fig. 1.1.

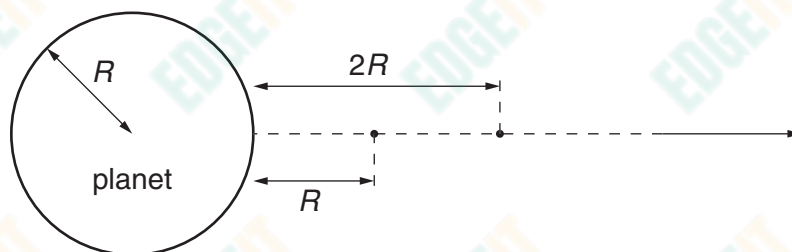


Fig. 1.1

The mass of the rocket, after its engines have been stopped, is  $m$ .

- (i) Show that, for the rocket to travel from a height  $R$  to a height  $2R$  above the planet's surface, the change  $\Delta E_p$  in the magnitude of the gravitational potential energy of the rocket is given by the expression

$$\Delta E_p = \frac{GMm}{6R}.$$

[2]

- (ii) During the ascent from a height  $R$  to a height  $2R$ , the speed of the rocket changes from  $7600 \text{ m s}^{-1}$  to  $7320 \text{ m s}^{-1}$ . Show that, in SI units, the change  $\Delta E_K$  in the kinetic energy of the rocket is given by the expression

$$\Delta E_K = (2.09 \times 10^6)m.$$

[1]

- (c) The planet has a radius of  $3.40 \times 10^6 \text{ m}$ .

- (i) Use the expressions in (b) to determine a value for the mass  $M$  of the planet.

$$M = \dots\dots\dots \text{ kg [2]}$$

- (ii) State one assumption made in the determination in (i).

.....  
 ..... [1]

- 2 (a) Use the kinetic theory of matter to explain why melting requires energy but there is no change in temperature.

.....

.....

.....

..... [3]

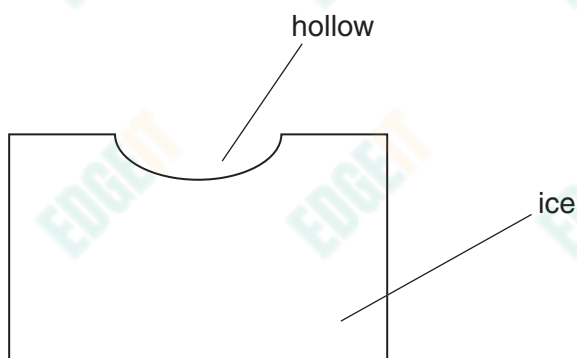
- (b) Define *specific latent heat of fusion*.

.....

.....

..... [2]

- (c) A block of ice at  $0^{\circ}\text{C}$  has a hollow in its top surface, as illustrated in Fig. 2.1.



**Fig. 2.1**

A mass of 160 g of water at  $100^{\circ}\text{C}$  is poured into the hollow. The water has specific heat capacity  $4.20\text{ kJ kg}^{-1}\text{ K}^{-1}$ . Some of the ice melts and the final mass of water in the hollow is 365 g.

- (i) Assuming no heat gain from the atmosphere, calculate a value, in  $\text{kJ kg}^{-1}$ , for the specific latent heat of fusion of ice.

specific latent heat = .....  $\text{kJ kg}^{-1}$  [3]

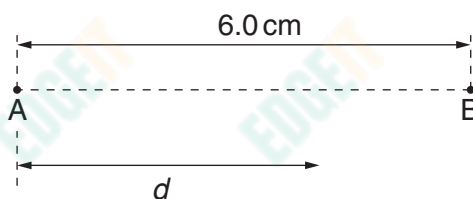
- (ii) In practice, heat is gained from the atmosphere during the experiment. This means that your answer to (i) is not the correct value for the specific latent heat. State and explain whether your value in (i) is greater or smaller than the correct value.

.....

.....

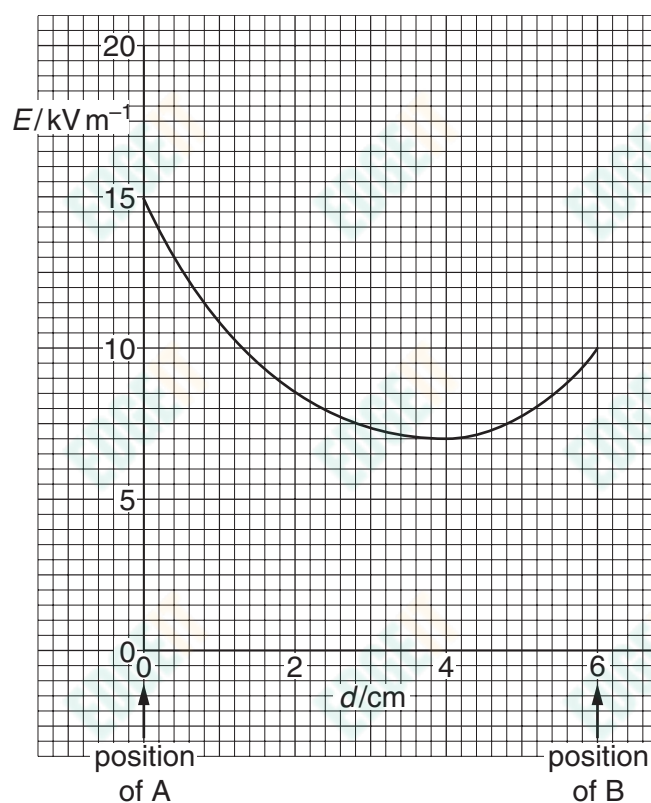
..... [2]

- 3 Two charged points A and B are separated by a distance of 6.0 cm, as shown in Fig. 3.1.



**Fig. 3.1**

The variation with distance  $d$  from A of the electric field strength  $E$  along the line AB is shown in Fig. 3.2.



**Fig. 3.2**

An electron is emitted with negligible speed from A and travels along AB.

- (a) State the relation between electric field strength  $E$  and potential  $V$ .

.....  
 ..... [2]

- (b) The area below the line of the graph of Fig. 3.2 represents the potential difference between A and B.

Use Fig. 3.2 to determine the potential difference between A and B.

potential difference = ..... V [4]

- (c) Use your answer to (b) to calculate the speed of the electron as it reaches point B.

speed = .....  $\text{ms}^{-1}$  [2]

- (d) (i) Use Fig. 3.2 to determine the value of  $d$  at which the electron has maximum acceleration.

$d$  = ..... cm [1]

- (ii) Without any further calculation, describe the variation with distance  $d$  of the acceleration of the electron.

.....  
 .....  
 ..... [2]

- 4 An ideal transformer has 5000 turns on its primary coil. It is to be used to convert a mains supply of 230V r.m.s. to an alternating voltage having a peak value of 9.0V.

(a) Calculate the number of turns on the secondary coil.

number = ..... [3]

- (b) The output from the transformer is to be full-wave rectified. Fig. 4.1 shows part of the rectifier circuit.

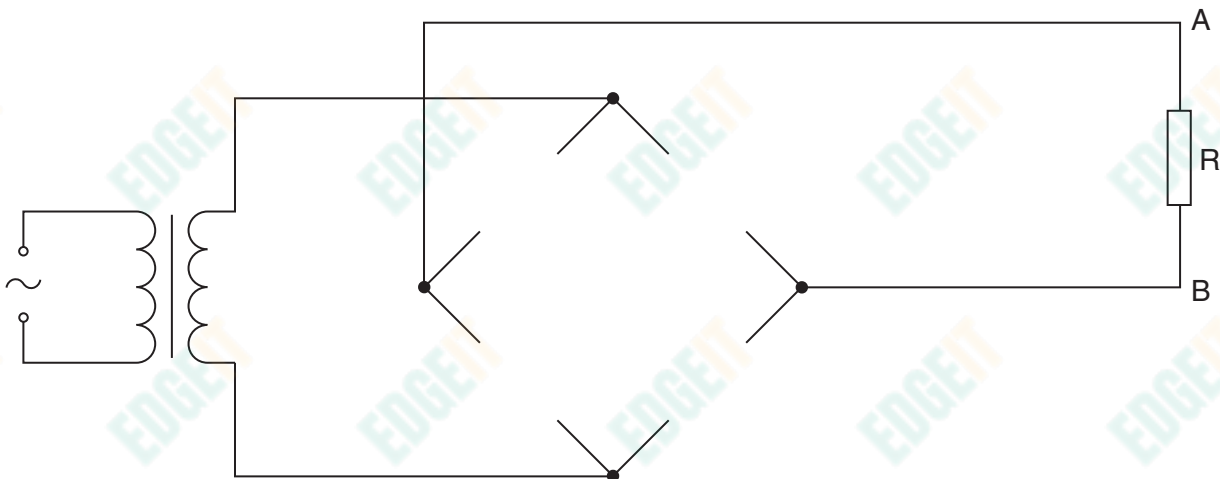


Fig. 4.1

On Fig. 4.1, draw

- (i) diode symbols to complete the diagram of the rectifier such that terminal A of the resistor R is positive with respect to terminal B, [2]
- (ii) the symbol for a capacitor connected to provide smoothing of the potential difference across the resistor R. [1]

- (c) Fig. 4.2 shows the variation with time  $t$  of the smoothed potential difference  $V$  across the resistor  $R$ .

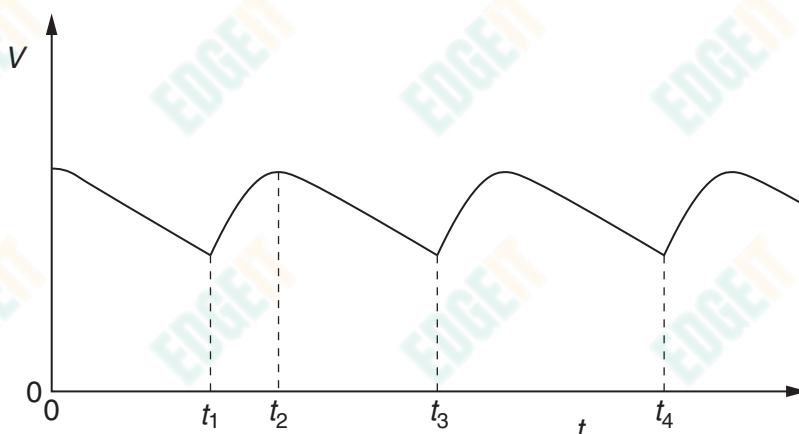


Fig. 4.2

- (i) State the interval of time during which the capacitor is being charged from the transformer.

from time ..... to time ..... [1]

- (ii) The resistance of the resistor  $R$  is doubled. On Fig. 4.2, sketch the variation with time  $t$  of the potential difference  $V$  across the resistor. [2]

- 5 (a) (i) Explain what is meant by a *photon*.

.....  
 ..... [1]

- (ii) Show that the photon energy of light of wavelength 350 nm is  $5.68 \times 10^{-19}$  J. [1]

- (iii) State the value of the ratio

$$\frac{\text{energy of photon of light of wavelength 700 nm}}{\text{energy of photon of light of wavelength 350 nm}}$$

ratio = ..... [1]

- (b) Two beams of monochromatic light have similar intensities. The light in one beam has wavelength 350 nm and the light in the other beam has wavelength 700 nm.

The two beams are incident separately on three different metal surfaces. The work function of each of these surfaces is shown in Fig. 5.1.

| metal     | work function / eV |
|-----------|--------------------|
| tungsten  | 4.49               |
| magnesium | 3.68               |
| potassium | 2.26               |

Fig. 5.1

- (i) Explain what is meant by the *work function* of the surface.

.....  
 .....  
 ..... [2]

- (ii) State which combination, if any, of monochromatic light and metal surface could give rise to photo-electric emission. Give a quantitative explanation of your answer.

.....

.....

.....

.....

.....

..... [3]

- 6 (a) Define the *decay constant* of a radioactive isotope.

.....  
.....  
..... [2]

- (b) Strontium-90 is a radioactive isotope having a half-life of 28.0 years. Strontium-90 has a density of  $2.54 \text{ g cm}^{-3}$ .

A sample of Strontium-90 has an activity of  $6.4 \times 10^9 \text{ Bq}$ . Calculate

- (i) the decay constant  $\lambda$ , in  $\text{s}^{-1}$ , of Strontium-90,

$$\lambda = \dots\dots\dots \text{s}^{-1} \quad [2]$$

- (ii) the mass of Strontium-90 in the sample,

$$\text{mass} = \dots\dots\dots \text{g} \quad [4]$$

(iii) the volume of the sample.

volume = ..... cm<sup>3</sup> [1]

(c) By reference to your answer in (b)(iii), suggest why dust that has been contaminated with Strontium-90 presents a serious health hazard.

.....  
.....  
..... [2]

- 7 A magnet is suspended vertically from a fixed point by means of a spring, as shown in Fig. 7.1.

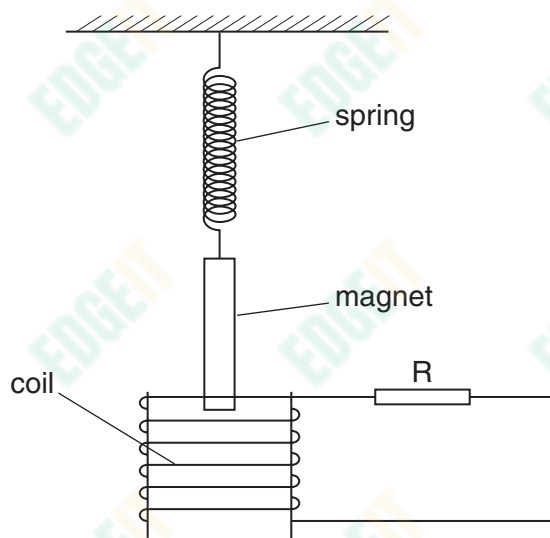


Fig. 7.1

One end of the magnet hangs inside a coil of wire. The coil is connected in series with a resistor  $R$ .

- (a) The magnet is displaced vertically a small distance  $D$  and then released. Fig. 7.2 shows the variation with time  $t$  of the vertical displacement  $d$  of the magnet from its equilibrium position.

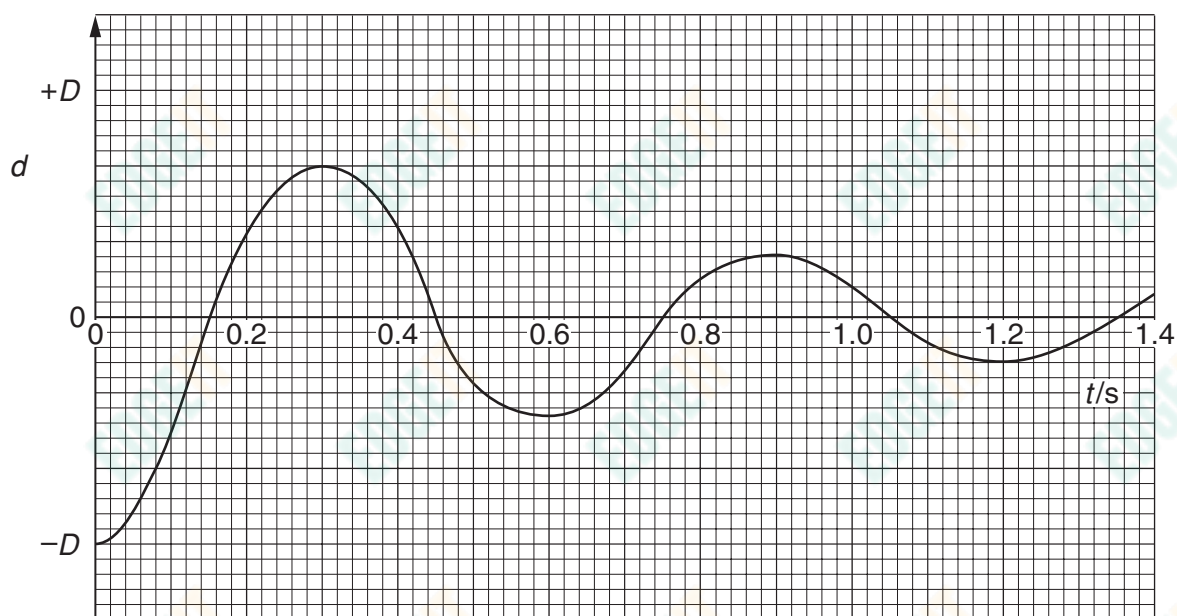


Fig. 7.2

- (i) State and explain, by reference to electromagnetic induction, the nature of the oscillations of the magnet.

.....

.....

.....

.....

.....

..... [5]

- (ii) Calculate the angular frequency  $\omega_0$  of the oscillations.

$$\omega_0 = \dots\dots\dots \text{ rad s}^{-1} \quad [2]$$

- (b) The resistance of the resistor R is increased.  
The magnet is again displaced a vertical distance  $D$  and released.  
On Fig. 7.2, sketch the variation with time  $t$  of the displacement  $d$  of the magnet. [2]

- (c) The resistor  $R$  in Fig. 7.1 is replaced by a variable-frequency signal generator of constant r.m.s. output voltage.  
The angular frequency  $\omega$  of the generator is gradually increased from about  $0.7\omega_0$  to about  $1.3\omega_0$ , where  $\omega_0$  is the angular frequency calculated in (a)(ii).
- (i) On the axes of Fig. 7.3, sketch a graph to show the variation with  $\omega$  of the amplitude  $A$  of the oscillations of the magnet. [2]

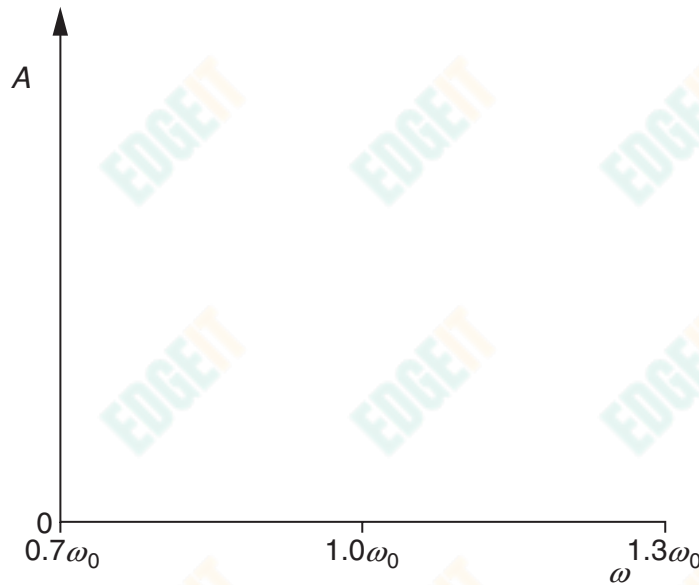


Fig. 7.3

- (ii) State the name of the phenomenon illustrated in the graph of Fig. 7.3.  
..... [1]
- (iii) Briefly describe one situation where the phenomenon named in (ii) is useful and one situation where it should be avoided.  
useful: .....  
.....  
avoid: .....  
..... [2]



## Section B

Answer **all** the questions in the spaces provided.

- 8 (a) State three characteristics of an ideal operational amplifier (op-amp).

1. ....
2. ....
3. .... [3]

- (b) An amplifier circuit for a microphone is shown in Fig. 8.1.

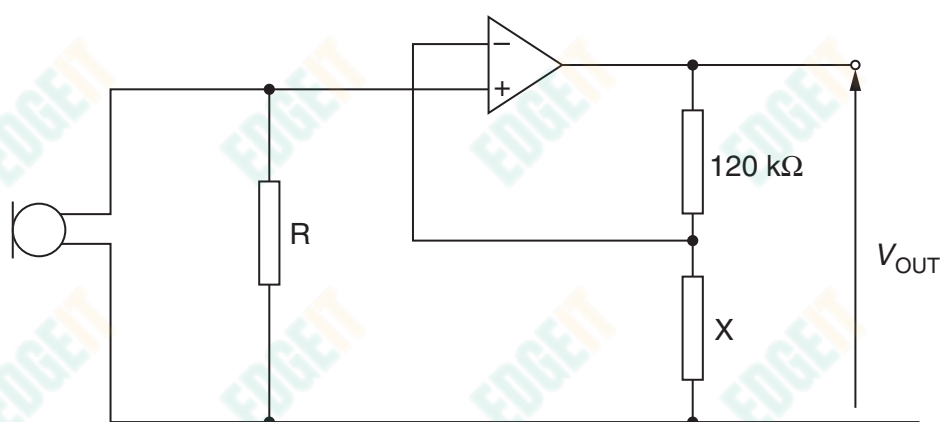


Fig. 8.1

- (i) Name the type of feedback used with this op-amp.

..... [1]

- (ii) The output potential difference  $V_{OUT}$  is 5.8V for a potential difference across the resistor R of 69mV. Calculate

1. the gain of the amplifier circuit,

gain = ..... [1]

2. the resistance of resistor X.

resistance = .....  $\Omega$  [2]

(iii) State one effect on the amplifier output of reducing the resistance of resistor X.

.....  
..... [1]

- 9 (a) Explain the principles behind the use of X-rays for imaging internal body structures.

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..... [4]

- (b) Describe how the image produced during CT scanning differs from that produced by X-ray imaging.

.....

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..... [5]

- 10 An analogue signal is sampled at a frequency of 5.0 kHz. Each sample is converted into a four-bit number and transmitted as a digital signal. Fig. 10.1 shows part of the digital signal.

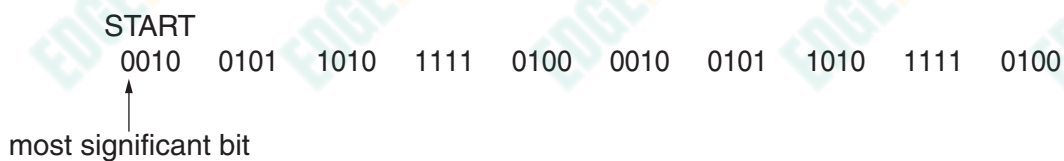


Fig. 10.1

The digital signal is transmitted and is finally converted into an analogue signal.

- (a) On the axes of Fig. 10.2, sketch a graph to show the variation with time  $t$  of this final analogue signal.

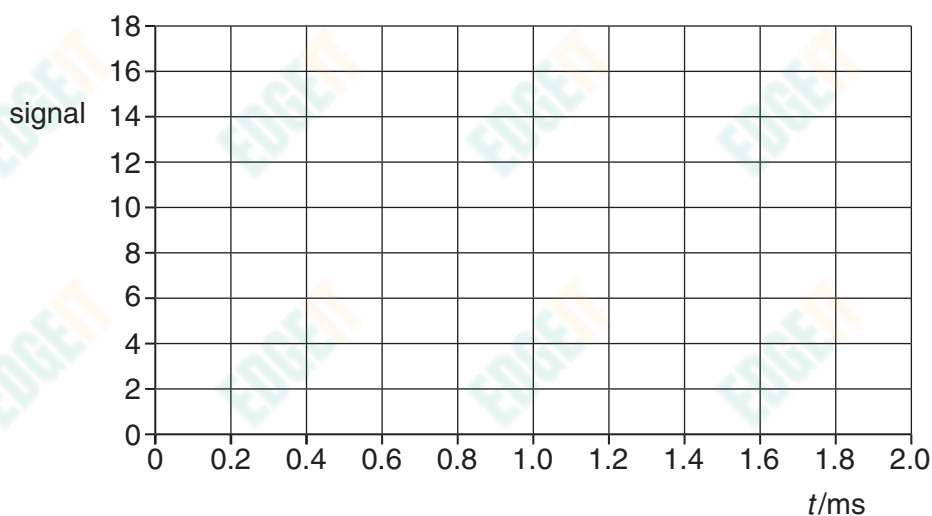


Fig. 10.2

[4]

- (b) Suggest two ways in which the reproduction of the original analogue signal could be improved.

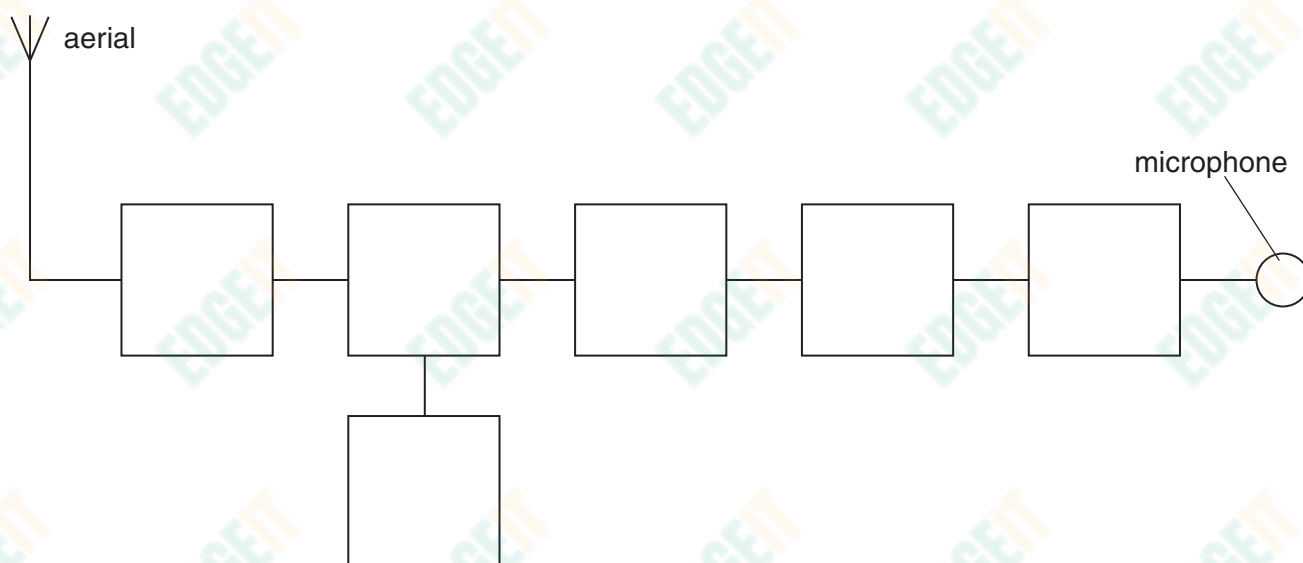
1. ....

.....

2. ....

..... [2]

- 11 (a)** Fig. 11.1 is a block diagram showing part of a mobile phone handset used for sending a signal to a base station.



**Fig. 11.1**

Complete Fig. 11.1 by labelling each of the blocks.

[3]

- (b)** Whilst making a call using a mobile phone fitted into a car, a motorist moves through several different cells. Explain how reception of signals to and from the mobile phone is maintained.

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[4]

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